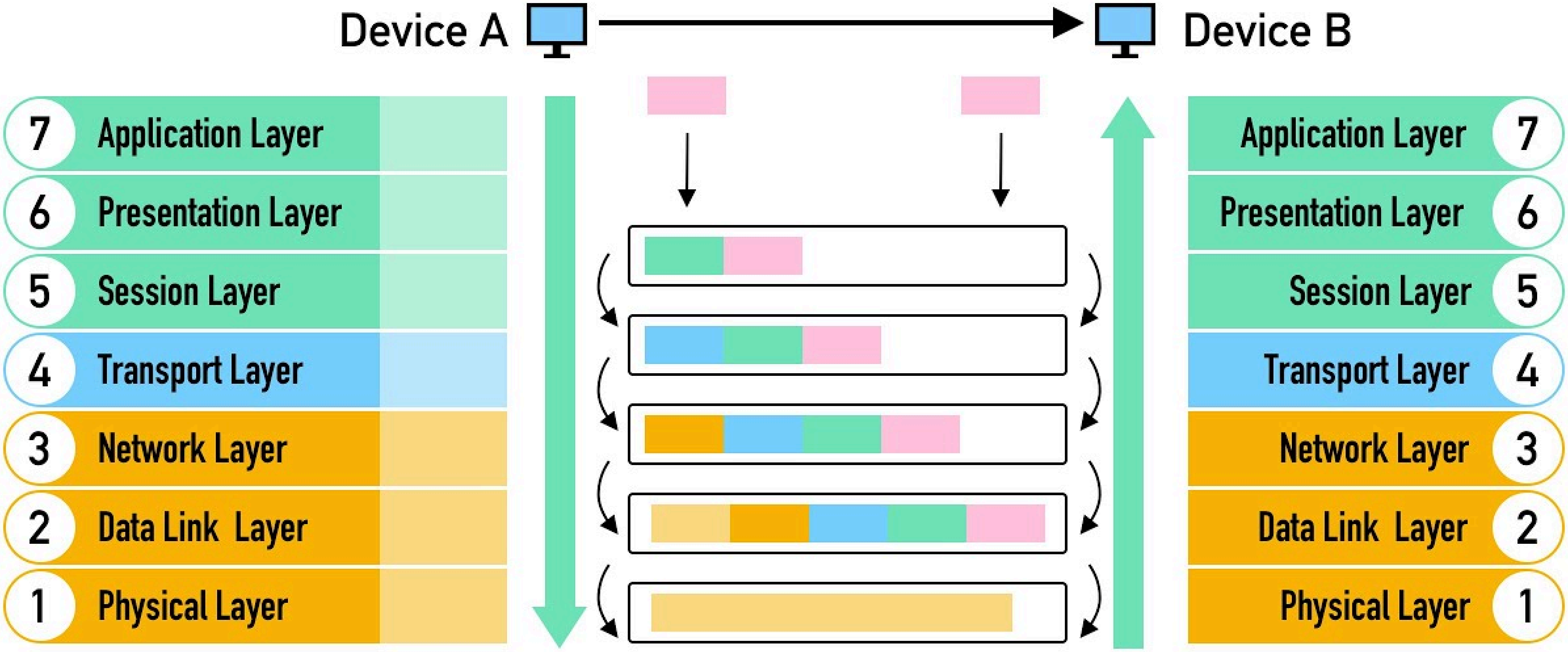
IT INFRASTRUCTURE AND COMPUTER NETWORKS

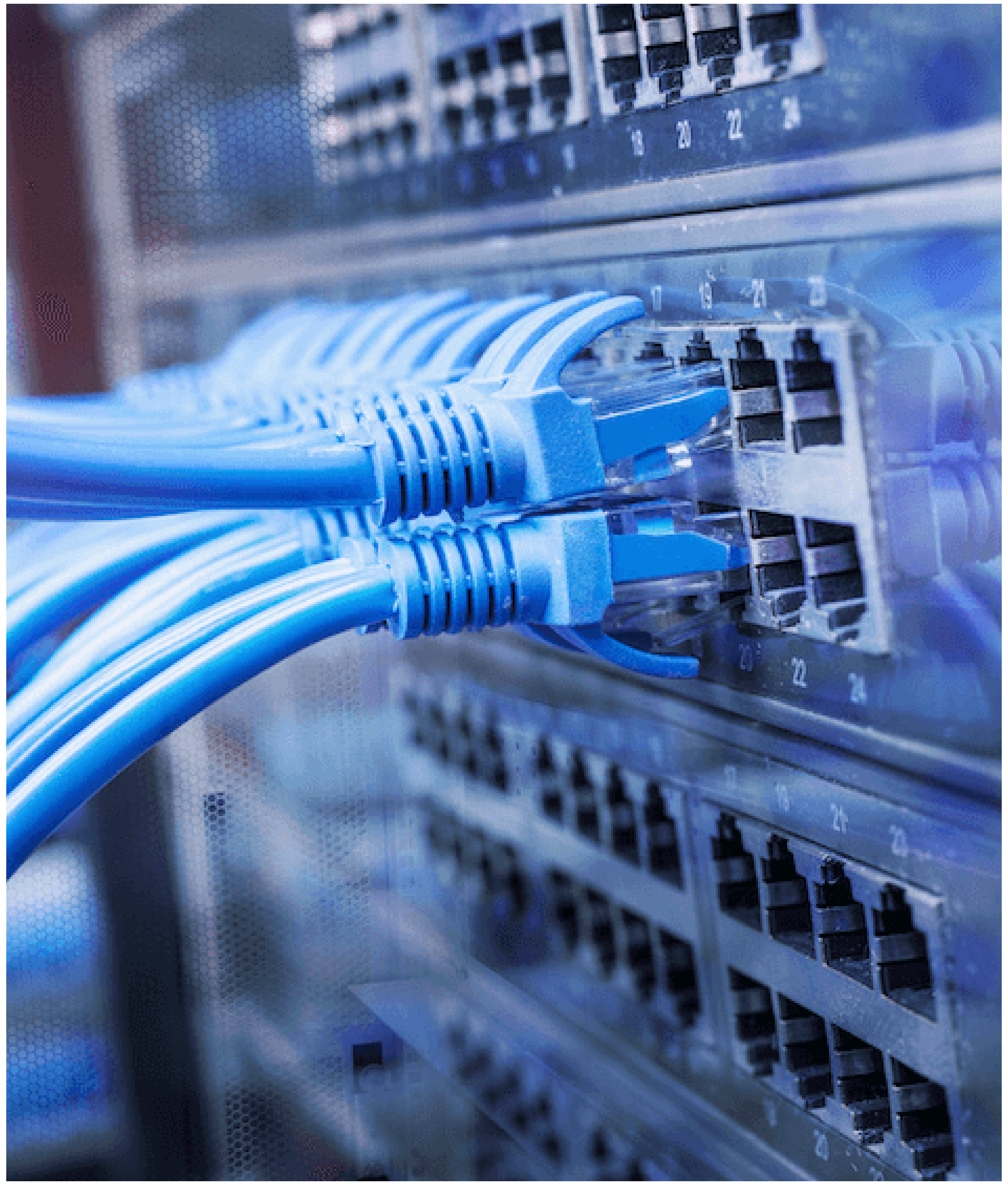
Week 3

What is OSI model?



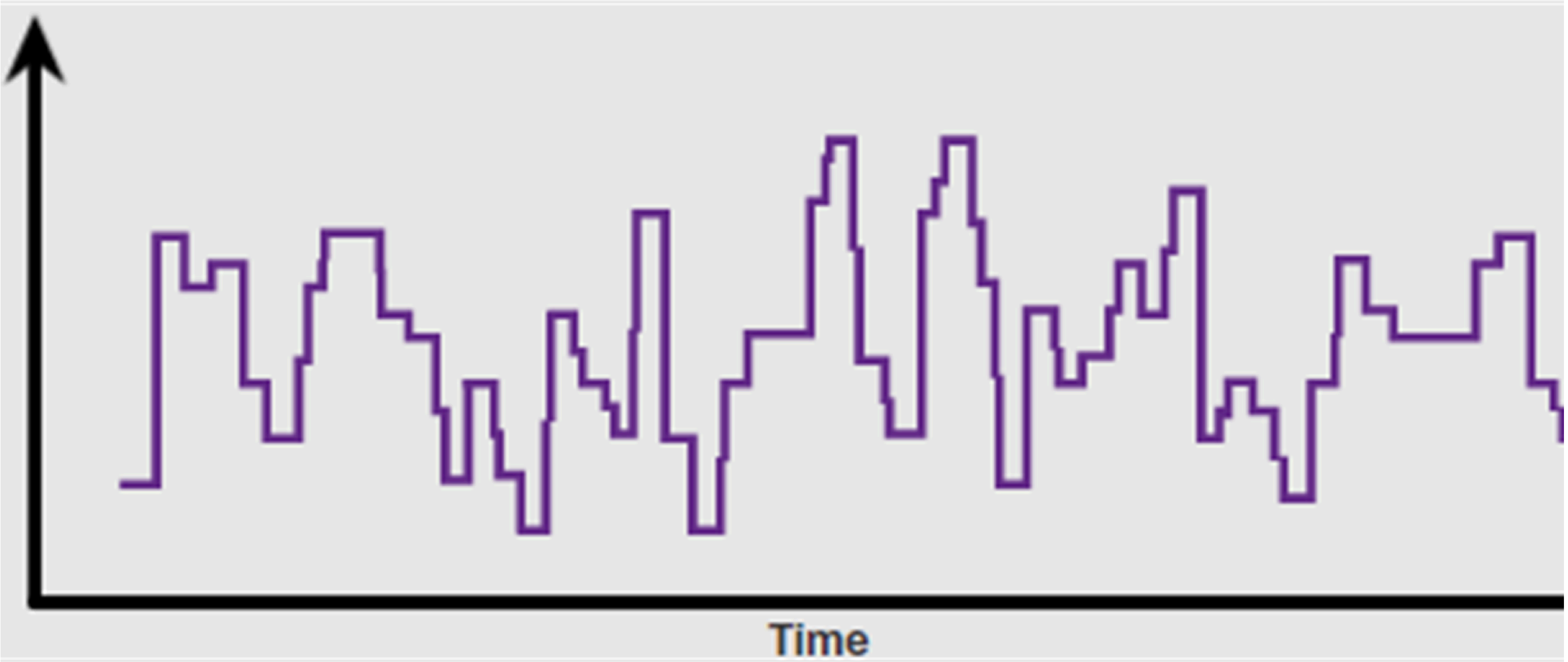
PHYSICAL LAYER

WAYS OF DATA TRANSMISSION

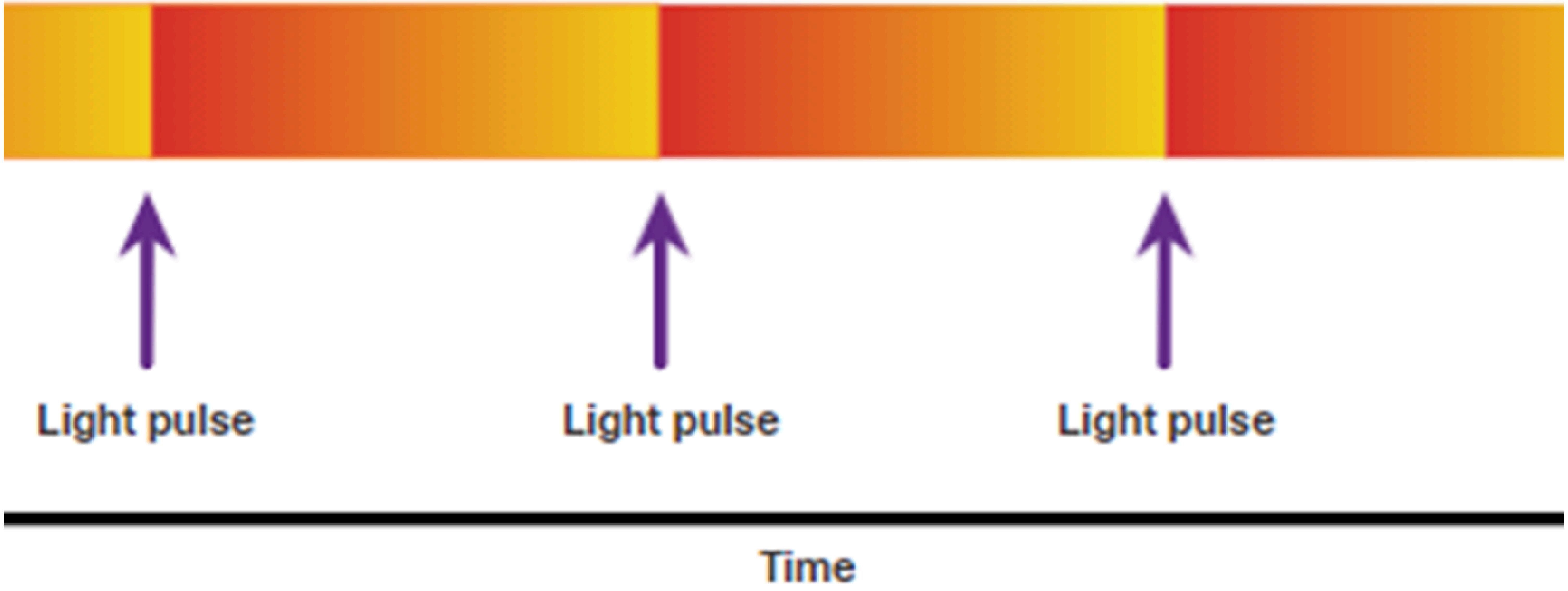


- Electrical signals
- Light pulses
- Microwave signals

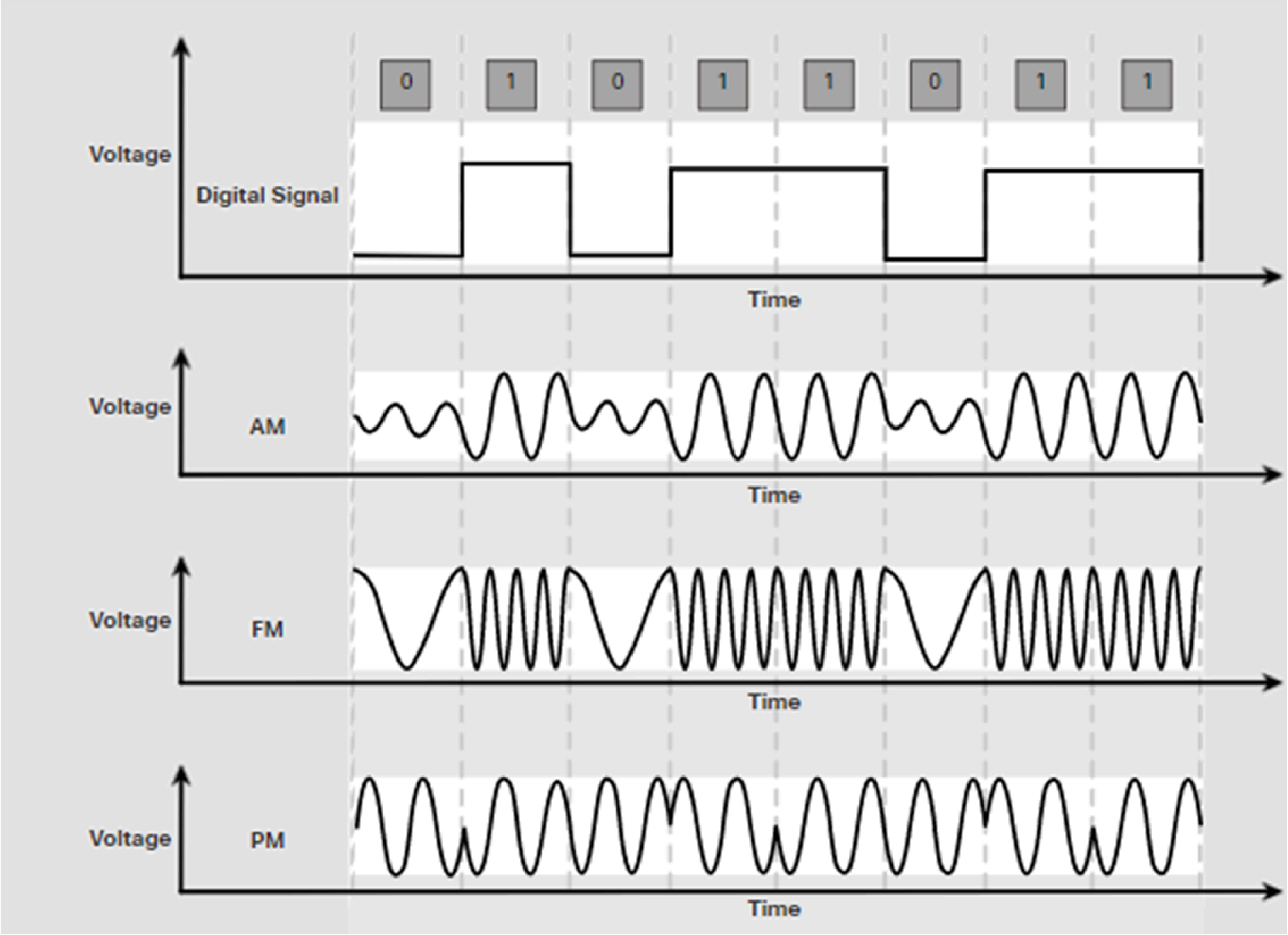
Electrical Signals Over Copper Cable



Light Pulses Over Fiber-Optic Cable



Microwave Signals Over Wireless



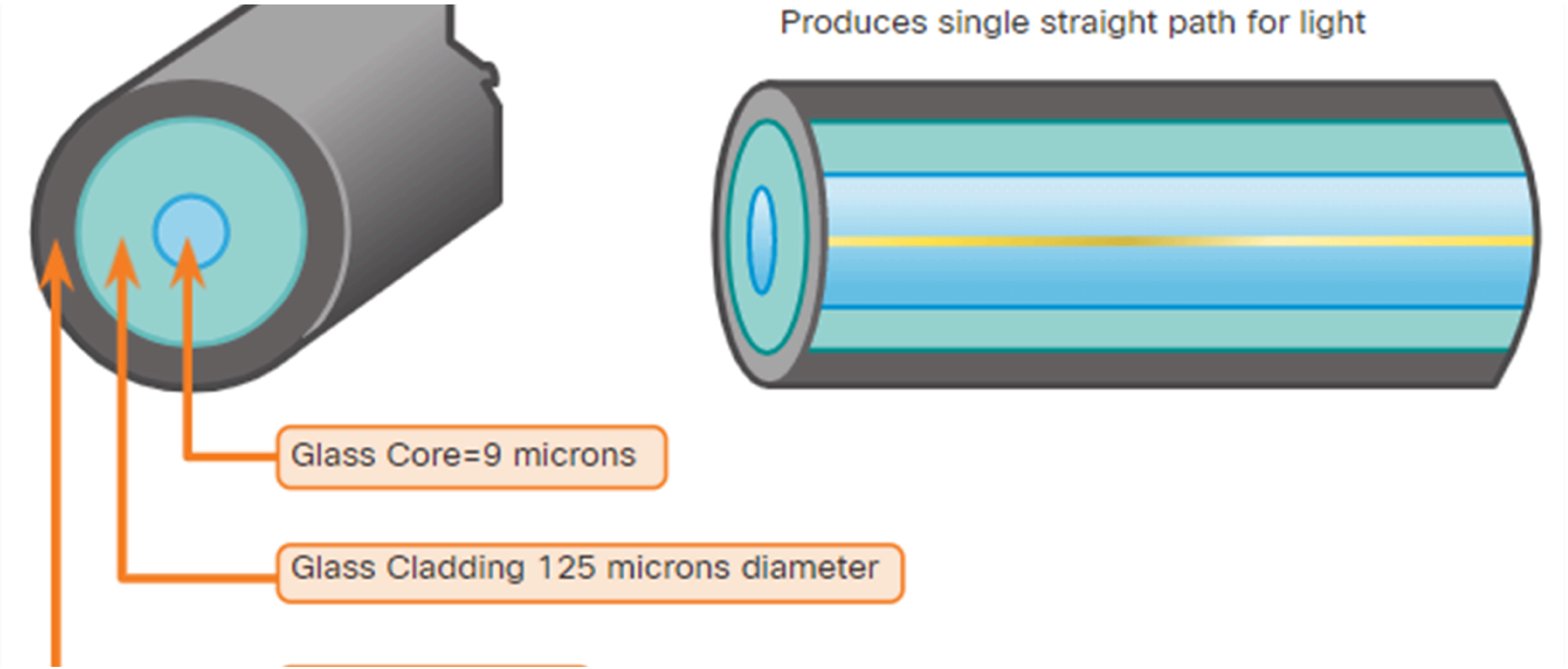
Types of Wireless Media

- Wi-Fi (IEEE 802.11)
- Bluetooth (IEEE 802.15)
- WiMAX (IEEE 802.16)
- Zigbee (IEEE 802.15.4)

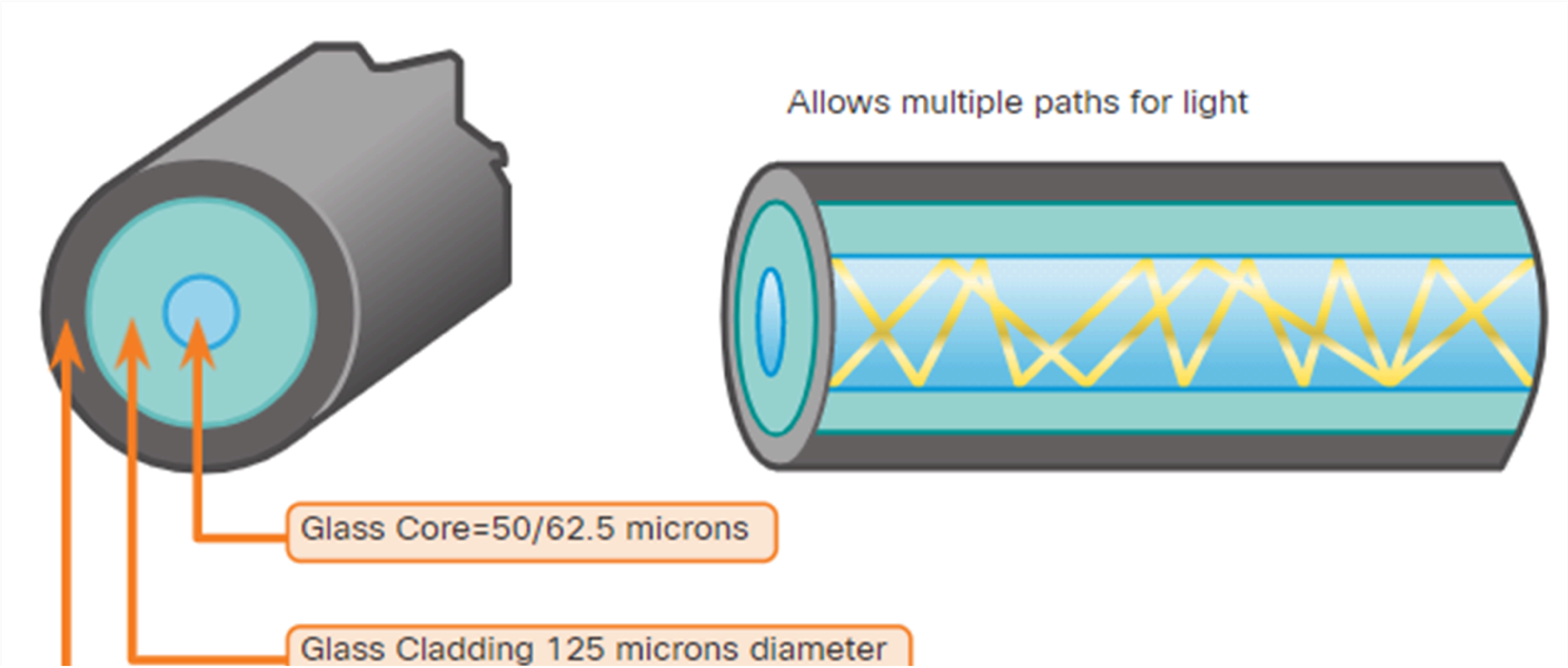
TYPES OF CABLES

- Copper cable
  - Unshielded twisted pair
  - Shielded twisted pair
  - Coaxial cable
- Optic cable

Single-Mode Fiber



Multimode Fiber



Types of Copper Cabling

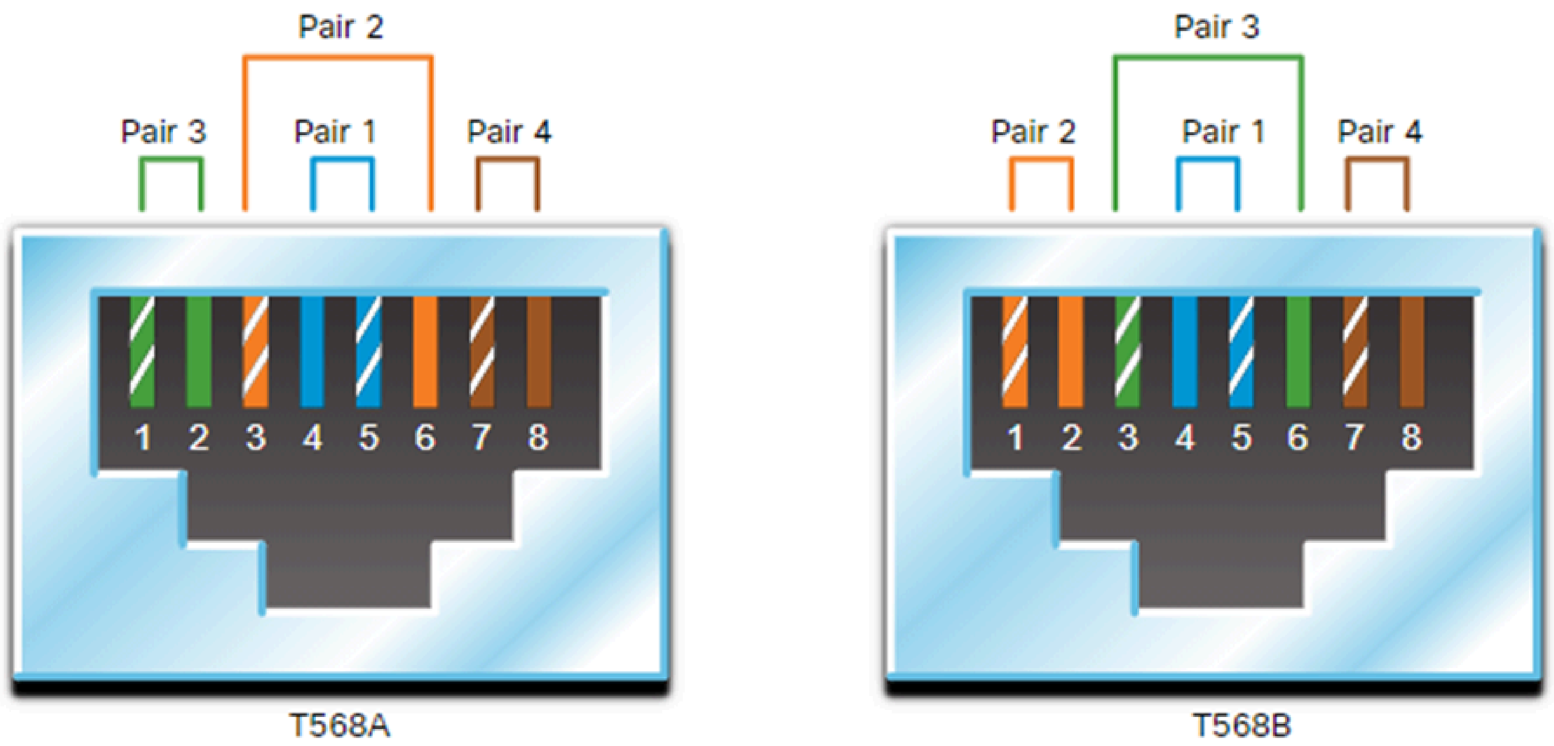
Unshielded Twisted-Pair (UTP) Cable

Shielded Twisted-Pair (STP) Cable

Coaxial Cable

UTP Cabling Connectors

RJ-45 UTP Plugs



RJ-45 UTP Sockets

Copper versus Fiber

Implementation Issues	UTP Cabling	Fiber-Optic Cabling
Bandwidth supported	10 Mb/s - 10 Gb/s	10 Mb/s - 100 Gb/s
Distance	Relatively short (1 - 100 meters)	Relatively long (1 - 100,000 meters)
Immunity to EMI and RFI	Low	High (Completely immune)
Immunity to electrical hazards	Low	High (Completely immune)
Media and connector costs	Lowest	Highest
Installation skills required	Lowest	Highest
Safety precautions	Lowest	Highest

Any questions?